

Reading and Sketching Slope Fields

Answer Key

AP Calculus AB/BC

Quick Answers

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| 1. (a) slope at the point = 3, — | 5. (a) y with zero slope = 0, — |
| 2. (a) slope at the point = 2, (b) y with zero slope = 0 | 6. (a) antiderivative of the x side = $\frac{x^2}{2}$, (b) value at $x = 2 = 2e^2$, — |
| 3. (a) slope at the point = $-\frac{3}{4}$, — | 7. (a) equilibrium values = 0, 4, — |
| 4. (a) general solution = x^2 , (b) constant of integration = 1, — | 8. (a) slope at the point = 1, (b) value at $x = 2 = 3.464$, — |
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What is verified

12 of 19 answers machine-checked against SymPy, across 8 problems.

— marks an answer only you can judge — problems 1, 3, 4, 5, 6, 7, 8. The worked solution states what a correct response must contain.

Curriculum

Level: AP Calculus AB/BC

FUN-7 – problems 1, 2, 3, 4, 5, 6, 7, 8

Difficulty 1–5 of 5 across 19 tagged checks.

1. Field for $\frac{dy}{dx} = x - y$, dot at $(0, 1)$. (a) Slope at $(2, -1)$. (b) Sketch the solution through the dot.

Solution (a). Substitute the point into the differential equation — that is all a slope field is, evaluated one point at a time: $\frac{dy}{dx} = 2 - (-1)$.

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Solution (b) — instructor-judged. The curve through $(0, 1)$ should start with slope $0 - 1 = -1$, dip, and then turn upward, staying tangent to every segment it passes. Award full credit when the sketch is tangent to the field throughout and passes through the dot; a curve that cuts across the segments earns nothing, however smooth it looks. (This family approaches the line $y = x - 1$ from above, which is a useful thing to point out.)

2. Field for $\frac{dy}{dx} = \frac{y}{2}$. (a) Slope at $(1, 4)$. (b) Where the segments are horizontal.

Solution (a). The slope does not depend on x at all: $\frac{dy}{dx} = \frac{4}{2}$.

$$\boxed{\frac{dy}{dx} = 2}$$

Solution (b). $\frac{y}{2} = 0$.

$$\boxed{y = 0}$$

Every segment along the horizontal axis is flat, and the whole row at a given height is parallel — the signature of a slope that depends only on y .

3. Field for $\frac{dy}{dx} = -\frac{x}{y}$ with $y > 0$, dot at $(3, 4)$. (a) Slope there. (b) Sketch and describe.

Solution (a). $-\frac{3}{4}$:

$$\boxed{-\frac{3}{4}}$$

Solution (b) — instructor-judged. The solution curves are the upper halves of circles centred at the origin; through $(3, 4)$ the curve is the arc of $x^2 + y^2 = 25$, rising to a flat top at $(0, 5)$ and falling away on both sides. Award full credit only when the sketch is tangent to the field *and* the description names the circular shape; one without the other is half.

4. Field for $\frac{dy}{dx} = 2x$, dot at $(1, 2)$. (a) General solution. (b) C . (c) Sketch.

Solution (a). Antidifferentiate the right-hand side:

$$\boxed{y = x^2 + C}$$

Solution (b). The curve passes through $(1, 2)$, so $2 = 1 + C$.

$$\boxed{C = 1}$$

Solution (c) — instructor-judged. The particular solution is $y = x^2 + 1$: an upward parabola with its vertex on the vertical axis at height 1, passing through the dot. Award full credit when the sketch is a parabola through the marked point that flattens where the printed segments are flat and steepens symmetrically to either side.

5. Field for $\frac{dy}{dx} = x$ or $\frac{dy}{dx} = y$. (a) Where $\frac{dy}{dx} = y$ is flat. (b) Which equation is printed.

Solution (a). $y = 0$:

$$\boxed{y = 0}$$

Solution (b) — instructor-judged. The printed field is $\frac{dy}{dx} = y$. In it the slope depends only on the height, so every segment in a horizontal row is parallel and the row along the horizontal axis is flat. Under $\frac{dy}{dx} = x$ the parallel segments would run down each *vertical* column instead, and the flat ones would lie along the vertical axis. Full credit needs that row-versus-column observation, not just the correct choice — a coin flip gets the choice right half the time.

6. Field for $\frac{dy}{dx} = xy$, dot at $(0, 2)$, solution $y = 2e^{x^2/2}$. (a) $\int x dx$. (b) y at $x = 2$. (c) Sketch.

Solution (a).

$$\boxed{\frac{x^2}{2} + C}$$

Separating gives $\ln|y| = \frac{x^2}{2} + C$, and exponentiating with $y(0) = 2$ produces the stated solution.

Solution (b). $y(2) = 2e^{4/2}$.

$$\boxed{2e^2}$$

About 14.78, which is off the top of the printed grid.

Solution (c) — instructor-judged. The curve leaves the dot flat (the slope at $x = 0$ is zero for every height), then climbs ever more steeply to the right and mirrors that to the left. Award full credit when the sketch is tangent to the field throughout, flat where the field is flat, and never crosses the horizontal axis.

7. Field for $\frac{dy}{dx} = y(4 - y)$, dot at $(0, 1)$. (a) Horizontal segments. (b) Long-run behaviour.

Solution (a). $y(4 - y) = 0$ at

$$\boxed{y = 0 \text{ or } y = 4}$$

These are the equilibrium solutions: constant functions whose graphs are the two flat rows in the field.

Solution (b) — instructor-judged. Between the two flat rows every segment slopes upward, so a solution starting at $y = 1$ rises. As it approaches $y = 4$ the segments flatten, so the curve levels off and approaches 4 from below without reaching it. Full credit needs both the rising behaviour and the levelling off at 4.

8. Field for $\frac{dy}{dx} = \frac{x+1}{y}$, $y > 0$, dot at $(0, 2)$, solution $y = \sqrt{x^2 + 2x + 4}$. (a) Slope at $(1, 2)$. (b) y at $x = 2$. (c) Sketch and agreement.

Solution (a). $\frac{1+1}{2}$:

$$\boxed{1}$$

Solution (b). $y(2) = \sqrt{4 + 4 + 4} = \sqrt{12}$.

$$\boxed{3.464}$$

Solution (c) — instructor-judged. The curve rises through $(0, 2)$, flattening near $x = -1$ where the numerator vanishes and steepening to the right. A correct response names one specific agreement: for example that the curve steepens as x grows because the slope grows with x , or that it flattens as the height grows because the slope is divided by y . Full credit needs a named feature, not the general claim that they agree.